Lecture Notes
EC226 Term 2 2024/25

Estimating Causal Effects

Readings:

- Stock and Watson (2003): Chapter 4 + 5
- Dougherty (2016): Chapter 2
- Wooldridge (2013): Chapter 2

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1 Selection Problem

Consider the problem consider by Angrist and Pischke (2009) of whether hospitals make people healthier. The National Health Interview Survey (NHIS) has two questions "During the past 12 months, was the respondent a patient in a hospital overnight?", and "Would you say your health in general is excellent, very good, good, fair, poor?" (measured on a scale 1 (excellent) to 5 (poor)) (2005 NHIS):

Group	Sample Size	Mean health status	Std. Error
Hospital	7774	2.79	0.014
No Hospital	90049	2.07	0.003

Now assuming we have a model:

$$Y_i = \alpha + \beta D_i + \varepsilon_i \tag{1}$$

where Y_i = response to health survey and $D_i = 1$ went to hospital, 0 otherwise. The OLS estimates of equation (1) would be

$$\hat{Y}_i = 2.07 + \underset{(49.2)}{0.72} D_i$$

with t-ratios in parentheses. This implies that going to hospital significantly worsens one's health!

But the problem is that the people who go to the hospital are almost certainly less healthy than those who do not go to hospital (so this is a problem of a non-random sample of people going to hospital).

What we want is:

$$\text{Potential Outcome} = \begin{cases} Y_{1i} \text{ if } \mathrm{D_i} = 1 \\ Y_{0i} \text{ if } \mathrm{D_i} = 0 \end{cases}$$

where Y_{0i} is the health status of the individual if they did not go to hospital and Y_{1i} is the health status of the individual if they did go to hospital and we want to measure $(Y_{1i} - Y_{0i})$ - the Average Treatment Effect (ATE). Unfortunately, we only observe:

$$Y_i = \begin{cases} Y_{1i} \text{ if } D_i = 1 \\ Y_{0i} \text{ if } D_i = 0 \end{cases} \Rightarrow Y_i = Y_{0i} + (Y_{1i} - Y_{0i}) D_i$$

So we must try and learn about the effects of hospitalization by comparing the average health of those who were and were not hospitalized.

$$\begin{aligned} \underbrace{E(Y_i|D_i = 1) - E(Y_i|D_i = 0)}_{\text{Observed difference}} = & \underbrace{E(Y_{1i}|D_i = 1) - E(Y_{0i}|D_i = 1)}_{\text{ATT (ave. treatment effect on treated)}} \\ & + \underbrace{E(Y_{0i}|D_i = 1) - E(Y_{0i}|D_i = 0)}_{\text{Selection bias}} \end{aligned}$$

We want the first term on the right hand side (as this is our best estimate of the ATE and is the ATT) of this equation (the health difference of the hospitalised, assuming we could observe them not having gone to hospital). The effect we observe also includes the selection bias term which is the difference in health if did not go to hospital between those who did and did not go to hospital. The bias arises in the regression above as we believe: $E(Y_{0i}|D_i = 1) - E(Y_{0i}|D_i = 0) > 0$.
So we need solutions to the bias in OLS estimation of equation (1).

2 Random Assignment

If D_i is randomly assigned then: $E(Y_{0i}|D_i = 1) = E(Y_{0i}|D_i = 0)$ and hence the selection bias is zero and:

$$\underbrace{E(Y_i|D_i = 1) - E(Y_i|D_i = 0)}_{\text{Observed difference}} = \underbrace{E(Y_{1i}|D_i = 1) - E(Y_{0i}|D_i = 1)}_{\text{ATT}}$$

In which case we can estimate equation (1) by OLS.

3 IV estimation

Find a suitable instrument(s) for D_i, assume this as a binary variable Z_i, (which is relevant and Exogenous ($\varepsilon_t \not\Rightarrow Z_t$ (e.g. Z is randomly assigned) and $Z_t \not\Rightarrow \varepsilon_t$ (Z only affects Y through D) in which case you can have the IV estimator written as:

$$b^{IV} = \frac{\overline{Y}_1 - \overline{Y}_0}{\overline{D}_1 - \overline{D}_0}$$

where $\overline{Y}_1$, $\overline{D}_1$ are the means of the appropriate variable when $Z = 1$ and $\overline{Y}_0$, $\overline{D}_0$ are the means of the appropriate variable when $Z = 0$.

In this case, there are a series of different types of individuals we have to consider, defined by the manner in which they react to the instrument:

	$Z = 0$	$Z = 1$
Compliers	Not treated	Treated
Always Takers	Treated	Treated
Never Takers	Not treated	Not treated
Defiers	Treated	Not treated

We assume that there are no Defiers (this is known as the monotonicity assumption). If one estimates the effect on all that are treated then we are likely to have a bias in the coefficient estimates as the never takers and the always takes are unlikely to be a random sample of the population.

So one can estimate the effect on Y of based on $Z = 0/1$, but this will give an *Intended to Treat* estimator, which is not the same as the ATT effect. The IV estimator (Local Average Treatment Effect – LATE) is based on only looking at the Compliers as there is no variation in Always Takers and Never Takers with the instrument.

Consider the example from last term: Y=Performance; D=Attend; and Z=9am class (when classes were allocated by the Department on a random basis rather than through a process whereby students select their own classes), then our individuals are:

	9am class	Not 9am class
Compliers	Miss class	Attend class
Always Takers	Attend class	Attend class
Never Takers	Miss class	Miss class
Defiers	Attend class	Miss class

4 Difference-in-Difference (DiD) Estimator

Consider general pooled cross-section model:

$$y_{it} = \alpha + d_t + \beta_1 x_{1it} + \beta_2 x_{2it} + \ldots + \beta_k x_{kit} + \varepsilon_{it} \qquad i = 1, \ldots N, \quad t = 1, \ldots T \tag{2}$$

where $d_t = \sum_{s=2}^{T} \delta_s D_{is}$ (are time dummies). Assume there are n_0 individuals in period $t = 0$, n_1 individuals in period $t = 1$ and ultimately n_T individuals in period $t = T$.

With pooled cross-section data a powerful estimation technique for estimating causal effects is difference in differences (DiD). Good DiD techniques rely on good natural experiments, such as a change of government policy. In a typical set-up there is: A treatment group: a set of observations which receive the new policy. A control group: a set of observations which does not receive the new policy. Both the treatment and control groups are observed both before and after the policy change.

The crucial assumption is the common trends assumption: in absence of the policy both the treatment and control groups would have followed the same trends. The greater the similarity between the treatment and control groups the more likely this assumption is to be satisfied.

Example: Card and Krueger (2000) observed that in February 1992 both New Jersey (NJ) and Pennsylvania (PA) (neighbouring states in the US) both charged a state minimum wage of 4.25 dollars. On April 1st 1992 NJ raised the state minimum wage from 4.25 to 5.05 dollars. The state minimum wage in PA remained at 4.25. Using data on number of people employed (among other things) in the same type of fast-food restaurants in NJ and PA before and after the policy change, where NJ is the treatment group and PA the control group.

This is a natural experiment: naturally occurring quasi-random variation in the minimum wage. The data was collected both before and after the policy change. There is no need for the restaurants to be the same in each period, as long as they are collected in the relevant geographic areas.

The estimating regression takes the form:

$$y_{ist} = \alpha + \gamma NJ_{ist} + \lambda d_{ist} + \delta \left(NJ_{ist} \times d_{ist} \right) + \varepsilon_{ist} \tag{3}$$

where: y_{ist} is FTE employment in restaurant i, in state, s, in period, t. With $s = 2$ (NJ; PA) and $T = 2$ (February, November). $NJ_{ist} = 1$ if the restaurant is in New Jersey (treatment state), and zero if in Pennsylvania (control state). $d_{ist} = 1$ for observations in November (post-treatment) and zero for observations in February (pre-treatment). $NJ_{ist} \times d_{ist}$ is an interaction term between the state and time dummies. With associated assumptions:

1. $E(\varepsilon_{ist}) = E(\varepsilon_{ist}|\mathbf{x}) = \mathbf{0}$, breaking down this assumption: $E(\varepsilon_{ijs}|NJ_{ist}) = 0$, $E(\varepsilon_{ist}|d_{ist}) = 0$, and importantly, $E(\varepsilon_{ist}|NJ_{ist} \times d_{ist}) = 0$ is the common trends assumption.

2. $V(\varepsilon_{ist}|\mathbf{x}) = \sigma^2$

3. $cov(\varepsilon_{ist}, \varepsilon_{jst}|\mathbf{x}) = \mathbf{0}$

4. $\varepsilon_{ist}|\mathbf{x} \sim N(0, \sigma^2)$

Turning dummy variables on and off we get:

- PA pre-treatment: $E(y_{ist}|NJ_{ist} = 0, d_{ist} = 0) = \alpha$
- NJ pre-treatment: $E(y_{ist}|NJ_{ist} = 1, d_{ist} = 0) = \alpha + \gamma$
- PA post-treatment: $E(y_{ist}|NJ_{ist} = 0, d_{ist} = 1) = \alpha + \lambda$
- NJ post treatment: $E(y_{ist}|NJ_{ist} = 1, d_{ist} = 1) = \alpha + \gamma + \lambda + \delta$

Such that the DiD estimate is: = (NJpost − NJpre) − (PApost − PApre) $= \delta$
Difference in differences estimate is calculated using average employment in fast food restaurants as given in the table below (standard errors in parentheses):

	PA	NJ	NJ-PA
FTE employment before	23.33	20.44	-2.89
	(1.35)	(0.41)	(1.44)
FTE employment after	21.17	21.03	-0.14
	(0.94)	(0.52)	(1.07)
Change in FTE	-2.16	0.59	**2.76**
	(1.25)	(0.54)	**(1.36)**

The DiD estimate is 2.76 with a standard error of 1.36 and the t-statistic is approximately 2, implying we reject no effect at the 5% level.
The estimated regression in the current example would be:

$$\hat{y}_{ist} = 23.33 + 2.89NJ_{ist} - 2.16d_{ist} + 2.76(NJ_{ist} \times d_{ist})$$

the DiD coefficient of 2.76 is significant at the 5% level. Diagrammatically we can see all of these coefficients on the following two diagrams.

Figure 1: Pictorial Difference-in-difference

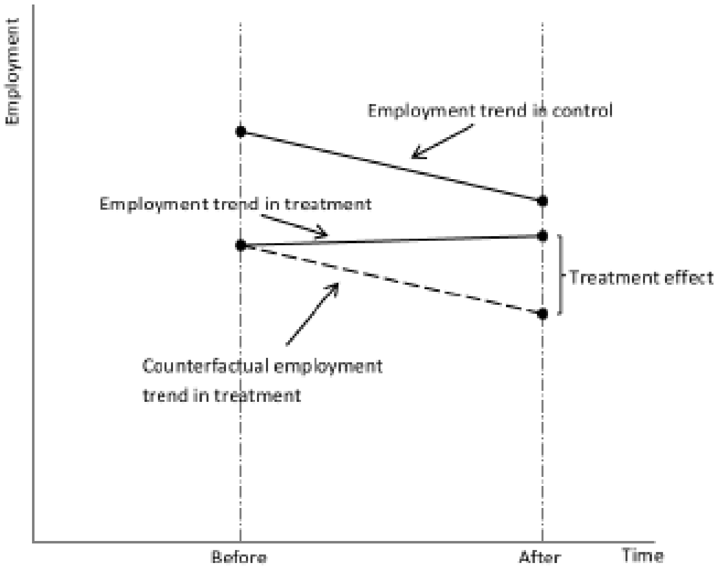
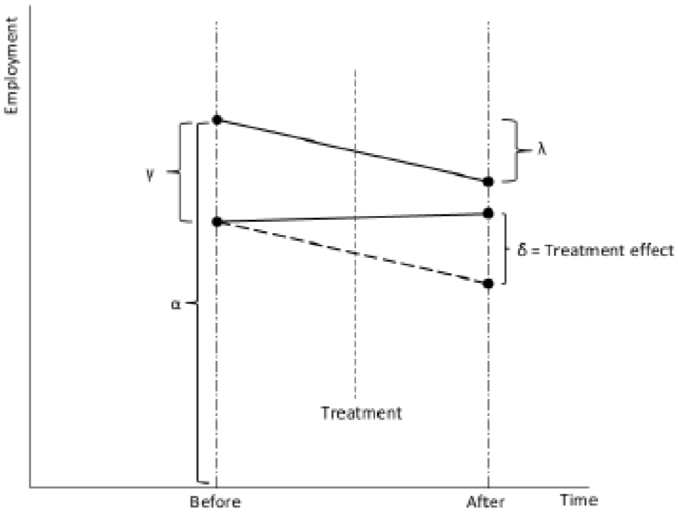


Figure 2: Pictorial representation of Equation x



The important assumption in all DiD strategies is the common trends assumption, which is that in the absence of the treatment, the treated and control groups would have followed a similar trend. The DiD estimate will have a causal interpretation if:

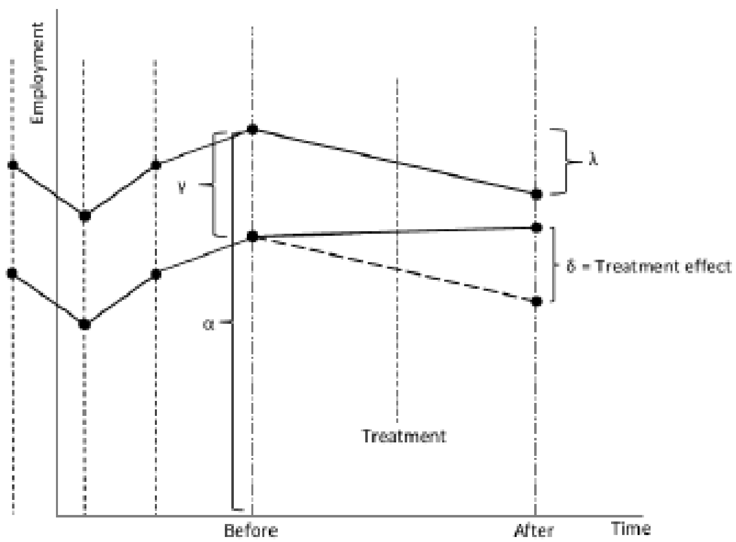
1. The employment trends would be the same in both states in the absence of the treatment (common trends assumption).

2. Treatment induces deviation from the trend as seen in the figures above. Note this is not the same as stating the control and treatment groups have the same level of the

outcome).[1]

This is a difficult assumption to test, as by definition you do not know what would have happened to the treatment group had they not entered the treatment.
Even if pre-treatment trends were the same (as below – common trends assumption), the concern is if anything else occurred between pre and post treatment in the treatment state but not the control. For instance, other policy changes which asymmetrically shifted average employment and this therefore might therefore require the presence of other control variables in the regression equation.

Figure 3: Common Trends Assumption



Alternatively one might think about extending the analysis. Extending the regression to more than two states and periods:

$$y_{ist}=\gamma_s + \lambda_t + \delta D_{ist} + \beta_1 x_{1ist} + \beta_2 x_{2ist} + ... + \beta_k x_{kist} + \varepsilon_{ist}$$

where $\gamma_s = \alpha + \sum_{j=2}^{s} \gamma_j S_{ijt}$; $\lambda_t = \sum_{j=2}^{T} \lambda_j d_{ist}$; $D_{ist} = 1$ if the treated state is in the treatment period, 0 otherwise. Notes on control group and control variables:

1. Selection of control group: the DiD set-up can be made much more general. For instance, instead of states one could use demographic groups, some of which are affected by a policy change and some are not.

2. Selection of control variables: it is important to only include exogenous controls (i.e. not affected by the policy), otherwise you will remove variation caused by the policy. That is, you will remove the effect you are trying to estimate.

[1] It is common to use regression to estimate DiD estimates since we can control for other variables and it is easy to calculate standard errors. We can extend the framework to include: i) multiple treatment/control states as well as time periods, ii) look at extended pre- and post-treatment effects

It is possible to extend the DiD idea to DiDiD. For instance, suppose:
USA State A: implements a health care policy, between $t = 0$ and $t = 1$, aimed at people 65 and over and health outcome is measured by some variable, Y.
USA State A: carry out a standard DiD analysis, using under 65's as the control group (maybe restricted people 60-65, say).
USA State A: $y_{it} = \alpha + \beta_1 O_{it} + \beta_2 d_t + \beta_3 (d_t \times O_{it}) + \varepsilon_{it}$,
where $O_{it} = 1$ if i is 65 or over, and zero otherwise, $d_t = 1$ if $t = 1$ and the final term is the interaction, such that β_3 gives the policy effect.
USA State A: The OLS estimator of β_3 will be unbiased if OLD and NOT OLD would have followed parallel health trends in absence of the policy.
USA State A: the problem with this is that other factors unrelated to the program might affect health in the younger generation relative to the elderly. Policy changes at the national level for instance.
Potential solution: extend analysis to include USA State B which did not implement a health care policy, but which would have been affected by policy changes at the national level.
USA State B: $y_{it} = \delta_0 + \delta_1 O_{it} + \delta_2 d_t + \delta_3 (d_t \times O_{it}) + \varepsilon_{it}$
where $O_{it} = 1$ if i is 65 or over, and zero otherwise, $d_t = 1$ if $t = 1$ and the final term is the interaction, such that δ_3 gives the policy effect in State B.
To get the DiDiD estimator:

$$\begin{aligned} y_{ijt} =& \delta_0 + \delta_1 O_{ijt} + \delta_2 d_{ijt} + \delta_3 d_{ijt} \times O_{ijt} + \gamma_0 S_{ijt} + \gamma_1 O_{ijt} \times S_{ijt} + \gamma_2 d_{ijt} \\ &+ \gamma_3 d_{ijt} \times O_{ijt} \times S_{ijt} + \varepsilon_{it} \end{aligned}$$

Where $S_{ijt} = 1$ if i is in State A and zero otherwise. In this situation γ_3, the term on the triple interaction, gives the DiDiD policy parameter.

- State=B, Control pre-treatment: $E(y_{ijt}|S_{ijt} = 0, O_{ijt} = 0, d_{ijt} = 0) = \delta_0$
- State=B, Treated pre-treatment: $E(y_{ijt}|S_{ijt} = 0, O_{ijt} = 1, d_{ijt} = 0) = \delta_0 + \delta_1$
- State=B, Control post-treatment: $E(y_{ijt}|S_{ijt} = 0, O_{ijt} = 0, d_{ijt} = 1) = \delta_0 + \delta_2$
- State=B, Treated post-treatment: $E(y_{ijt}|S_{ijt} = 0, O_{ijt} = 1, d_{ijt} = 1) = \delta_0 + \delta_1 + \delta_2 + \delta_3$

In which case:

$$\begin{aligned} \delta_3 =& [E(y_{ijt}|S_{ijt} = 0, O_{ijt} = 1, d_{ijt} = 1) - E(y_{ijt}|S_{ijt} = 0, O_{ijt} = 0, d_{ijt} = 1)] \\ &- [E(y_{ijt}|S_{ijt} = 0, O_{ijt} = 1, d_{ijt} = 0) - E(y_{ijt}|S_{ijt} = 0, O_{ijt} = 0, d_{ijt} = 0)] \end{aligned}$$

- State=A, Control pre-treatment: $E(y_{ijt}|S_{ijt} = 1, O_{ijt} = 0, d_{ijt} = 0) = \delta_0 + \gamma_0$
- State=A, Treated pre-treatment: $E(y_{ijt}|S_{itt} = 1, O_{itt} = 1, d_{itt} = 0) = \delta_0 + \delta_1 + \gamma_0 + \gamma_1$
- State=A, Control post-treatment: $E(y_{ijt}|S_{ijt} = 1, O_{ijt} = 0, d_{ijt} = 1) = \delta_0 + \delta_2 + \gamma_0 + \gamma_2$

- State=A, Treated post-treatment: $E(y_{ijt}|S_{ijt}=1,O_{ijt}=1,d_{ijt}=1)=\delta_0+\delta_1+\delta_2+\delta_3+\gamma_0+\gamma_1+\gamma_2+\gamma_3$

In which case:

$$\begin{aligned}\delta_3+\gamma_3=&[E(y_{ijt}|S_{ijt}=1,O_{ijt}=1,d_{ijt}=1)-E(y_{ijt}|S_{ijt}=1,O_{ijt}=0,d_{ijt}=1)]\\&-[E(y_{ijt}|S_{ijt}=1,O_{ijt}=1,d_{ijt}=0)-E(y_{ijt}|S_{ist}=1,O_{ijt}=0,d_{ijt}=0)]\end{aligned}$$

Such that the DiD estimate is:

$$\gamma_3=[(\bar{y}_{A;O;1}-\bar{y}_{A;O;0})-(\bar{y}_{A;NO;1}-\bar{y}_{A;NO;0})]-[(\bar{y}_{B;O;1}-\bar{y}_{B;O;0})-(\bar{y}_{B;NO;1}-\bar{y}_{B;NO;0})]$$

Where, State A is the policy state and B is the non-policy state, O is old and NO is not old, and 1 is period after policy and 0 is period before policy.
The DiDiD estimator has 4 terms:

1. $(\bar{y}_{A;O;1}-\bar{y}_{A;O;0})$: Difference in mean health in OLD between t = 1 and t = 0 in State A.
2. $(\bar{y}_{A;NO;1}-\bar{y}_{A;NO;0})$: Difference in mean health in NOT OLD between t = 0 and t = 1 in State A.
3. $(\bar{y}_{B;O;1}-\bar{y}_{B;O;0})$: Difference in mean health in OLD between t = 0 and t = 1 in State B.
4. $(\bar{y}_{B;NO;1}-\bar{y}_{B;NO;0})$: Difference in mean health in NOT OLD between t = 0 and t = 1 in State B.

5 Propensity Score Matching (PSM)

PSM are methods use to try and balance the treated group and the control group, by trying to model the treatment. There are many example of people using PSM methods see, for example, Nguyen et al. (2006) (please note as with most things there are many bad examples of inappropriate use of PSM methdos).

Consider the example in which we are looking at the effect of introducing clinic in villages on infant mortality. We have data on 9 villages and the data is below:

T	*imrate*	*povrate*	*pcdocs*
1	10	0.5	0.1
1	15	0.6	0.2
1	22	0.7	0.1
1	19	0.6	0.2
0	25	0.6	0.1
0	19	0.5	0.2
0	4	0.1	0.4
0	8	0.3	0.5
0	6	0.2	0.4

Where $T = 1$ if has a clinic and 0 otherwise *imrate* is the infant mortality rate (per 1000 births), *povrate* is a measure of poverty and *pcdocs* is number of doctors per capita. If we test for a difference in means we have:

```
. ttest imrate, by(t)

Two-sample t test with equal variances
------------------------------------------------------------------------------
   Group |     Obs        Mean    Std. Err.   Std. Dev.   [95% Conf. Interval]
---------+--------------------------------------------------------------------
       0 |       5        12.4    4.081666    9.126883    1.067478    23.73252
       1 |       4        16.5    2.598076    5.196152    8.231762    24.76824
---------+--------------------------------------------------------------------
combined |       9    14.22222    2.504317    7.512952    8.447256    19.99719
---------+--------------------------------------------------------------------
    diff |                -4.1    5.160149               -16.30181    8.101815
------------------------------------------------------------------------------
    diff = mean(0) - mean(1)                                      t =  -0.7946
Ho: diff = 0                                     degrees of freedom =        7

    Ha: diff < 0                 Ha: diff != 0                 Ha: diff > 0
 Pr(T < t) = 0.2265         Pr(|T| > |t|) = 0.4530          Pr(T > t) = 0.7735
```

In which we find that having a clinic is actually increasing the infant mortality by 4.1 per 1000 births. However, presumably clinics were put in those villages that are poorer and we would expect to have a higher mortality rate. If we observed data before the clinics were introduced we could do a diff-in-diff model and obtain the true effects of the clinic. In the absence of this an alternative is to attempt to construct the model for treatment and then use this to create a modified control group which looks much more like the treated group

and then use this reconstructed control group to compare with the treated group.
PSM essentially has two steps:

1. Estimate a probit/logit model of
$P(T = 1) = F(\beta_0 + \beta_1 x_1 + \ldots + \beta_k x_k)$
and use the predicted probabilities to match cases from the treat group to similar cases in the control group.

2. Do a difference in means test between the treated group and the artificially constructed control group.

In our example we are using psmatch2 in Stata:

```
. psmatch2 t povrate pcdocs, out(imrate)

Probit regression                                 Number of obs   =          9
                                                  LR chi2(2)      =       5.44
                                                  Prob > chi2     =     0.0659
Log likelihood =  -3.462284                       Pseudo R2       =     0.4400

------------------------------------------------------------------------------
           t |      Coef.   Std. Err.      z    P>|z|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
     povrate |   8.910143   8.326615     1.07   0.285    -7.409723    25.23001
      pcdocs |  -.0294659   8.185052    -0.00   0.997    -16.07187    16.01294
       _cons |  -4.692824   5.508573    -0.85   0.394    -15.48943     6.10378
------------------------------------------------------------------------------
----------------------------------------------------------------------------------------
   Variable     Sample |    Treated     Controls   Difference         S.E.   T-stat
----------------------------+-----------------------------------------------------------
     imrate  Unmatched |       16.5         12.4          4.1    5.1601495     0.79
                   ATT |       16.5         23.5           -7   4.24264069    -1.65
----------------------------+-----------------------------------------------------------
Note: S.E. does not take into account that the propensity score is estimated.

           | psmatch2:
 psmatch2: |   Common
 Treatment |  support
assignment | On suppor |     Total
-----------+-----------+----------
 Untreated |         5 |         5
   Treated |         4 |         4
-----------+-----------+----------
     Total |         9 |         9
```

And we see a significant negative effect of the treatment (ATT=-7, although marginally insignificant at 5% level), compared to what we saw previously which was a positive 4.1 effect.
It might be important to compare the means of the explanatory variables for the treated and the new control group of the explanatory variable (covariates) to check balance betwwen the two groups (in our case we see no significant difference based on povrate and pcdocs).
In Stata you can do this as:

```
. pstest povrate pcdocs

-----------------------------------------------------------------------------
                        |       Mean               |     t-test    |  V(T)/
Variable                | Treated Control    %bias |    t    p>|t| |  V(C)
------------------------+--------------------------+---------------+----------
povrate                 |      .6     .575    15.9 |   0.52  0.620 |  2.67
pcdocs                  |     .15     .125    20.3 |   0.65  0.537 |  1.33
-----------------------------------------------------------------------------
* if variance ratio outside [0.06; 15.44]

---------------------------------------------------------------------
Ps R2   LR chi2   p>chi2   MeanBias   MedBias      B       R     %Var
---------------------------------------------------------------------
0.109      1.21    0.546     18.1       18.1      72.7*   24.84*     0
---------------------------------------------------------------------
* if B>25%, R outside [0.5; 2]
```

An alternative estimator uses the teffects command in Stata, from which we get:

```
. teffects psmatch (imrate) (t povrate pcdocs, probit)
note: variance correction results in a negative variance estimate; ignoring the
correction term

Treatment-effects estimation                   Number of obs      =          9
Estimator      : propensity-score matching     Matches: requested =          1
Outcome model  : matching                                     min =          1
Treatment model: probit                                       max =          2
------------------------------------------------------------------------------
             |              AI Robust
      imrate |      Coef.   Std. Err.      z    P>|z|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
ATE          |
           t |
   (1 vs 0)  |  -3.666667   3.479511    -1.05   0.292    -10.48638    3.153049
------------------------------------------------------------------------------
```

In this case the estimate is negative, but the effect is insignificant. To test the extent to which there is balance between the treatment and control group:

```
. tebalance summarize
note: refitting the model using the generate() option

  Covariate balance summary
                                                Raw      Matched
                          -----------------------------------------
                          Number of obs =          9           18
                          Treated obs   =          4            9
                          Control obs   =          5            9
                          -----------------------------------------

  -----------------------------------------------------------------
                  |Standardized differences          Variance ratio
                  |        Raw     Matched           Raw    Matched
  ----------------+------------------------------------------------
          povrate |   1.649893    .7572824      .1550387   .1397059
           pcdocs |  -1.380397   -.8528029      .1234568         .1
  -----------------------------------------------------------------
```

Where you want the Matched differences to be close to zero and the Matched Variance Ratio to be close to unity.

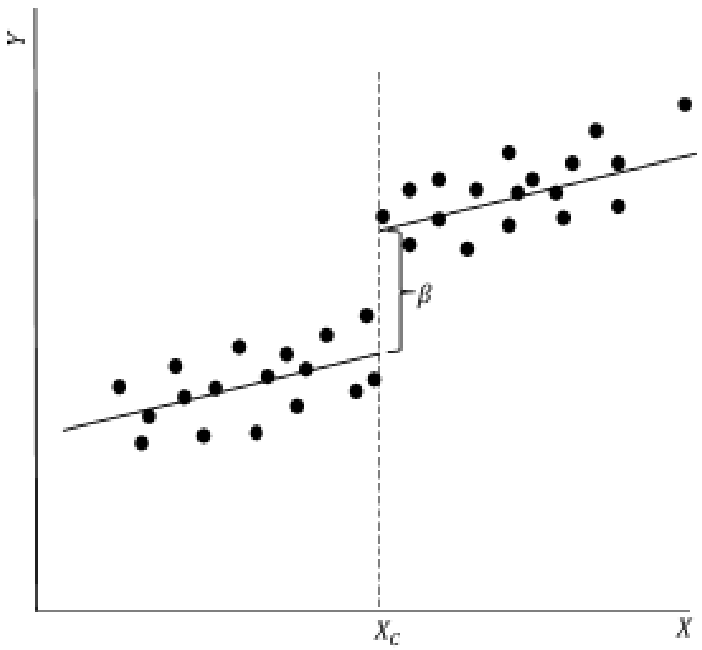
6 Regression Discontinuity Design (RDD)

RDDs are examples of quasi-experimental designs to estimate causal effects, there are many examples of RDDs in the literature, see for example Mealli and Rampichini (2012). The basic principle behind RDD is to use an arbitrary rule in an attempt to satisfy the assumption: $E(\varepsilon_i|\mathbf{x}) = 0$. For example,

Question: What is the impact of achieving a merit in a math test (during the year) on a child's final math exam score (at the end of the year)? A merit is awarded for a mark ≥ 70. The final exam score is measured between (0, 100). Intuitively: comparing the final exam scores, between children just above and below the 70 boundary, will give the causal impact of achieving a merit. Under the assumption that children above and below 70 are similar in all other respects.

The general set-up is the following: We are interested in estimating the impact of some treatment (D) on some outcome variable (y). You observe a continuous variable, x say, for a group of individuals. There is a cut-off along x, X^C say, where if $x \geq X^C$ the individual receives the treatment, such that $D = 1$. If $x < X^C$ the individual does not receive the treatment, such that $D = 0$ (see Figure 4).

Figure 4: Linear Regression Discontinuity



The causal parameter can be estimated using a regression of the form:

$$y_i = \alpha + \delta x_i + \beta D_i + \varepsilon_i$$

where y_i is the outcome variable; x_i is known as the assignment variable (or forcing variable or running variable); X^C is the cut-off along the assignment variable which assigns the treatment; D_i is a dummy variable equal to one if $x_i \geq X^C$ and zero if $x_i < X^C$; and ε_i is the error term. The model can be estimated using OLS and will give the causal estimate if $E(\varepsilon_i|D_i, x_i) = 0$.

NOTE: the closer the observations are to the cut-off the greater our trust in the causal estimate is likely to be, however, we are often restricted in terms of sample size. In the above diagram is clear that:

1. You do not observe treated and control individuals at the same level of x.
2. As such the causal (treatment) effect is based on extending (extrapolating) the regression function.

For this reason it is very important to try different specifications of the regression function. The above graph is an example of a linear assignment mechanism. Another well used mechanisms tested are higher order polynomial forms.

The causal parameter can be estimated using a regression of the form:

$$y_i = \alpha + \delta_1 x_i + \delta_2 x_i^2 + \delta_3 x_i^3 + \beta D_i + \varepsilon_i$$

Where: y_i is the outcome variable; (x_i, x_i^2, x_i^3) is known as the assignment variable (and is now a 3rd order polynomial); X^C is the cut-off along the assignment variable which assigns the treatment; D_i is a dummy variable equal to one if $x_i \geq X^C$ and zero if $x_i < X^C$; ε_i is the error term (see Figure 5). The model can be estimated using OLS and will give the causal estimate if $E(\varepsilon_i|D_i, x_i, x_i^2, x_i^3) = 0$.

NOTE: there is no need for control variables as D_i is random; however, controls may improve precision.

Figure 5: 3rd Order Polynomial Regression Discontinuity

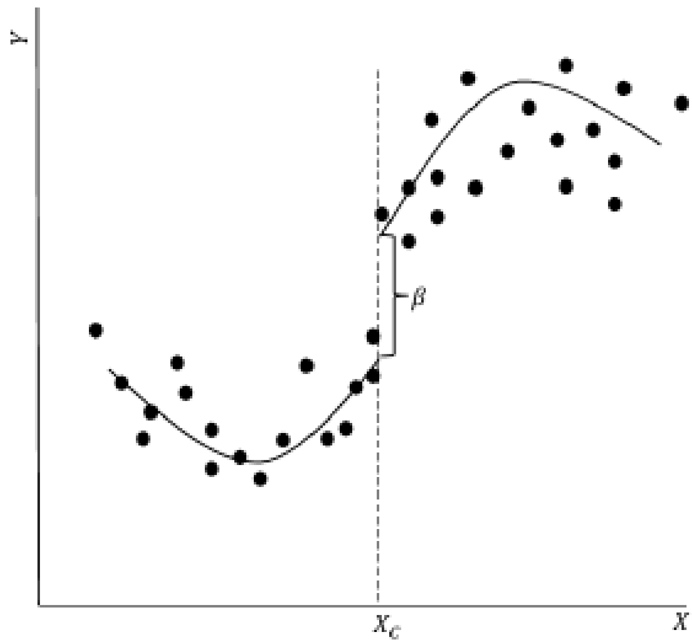
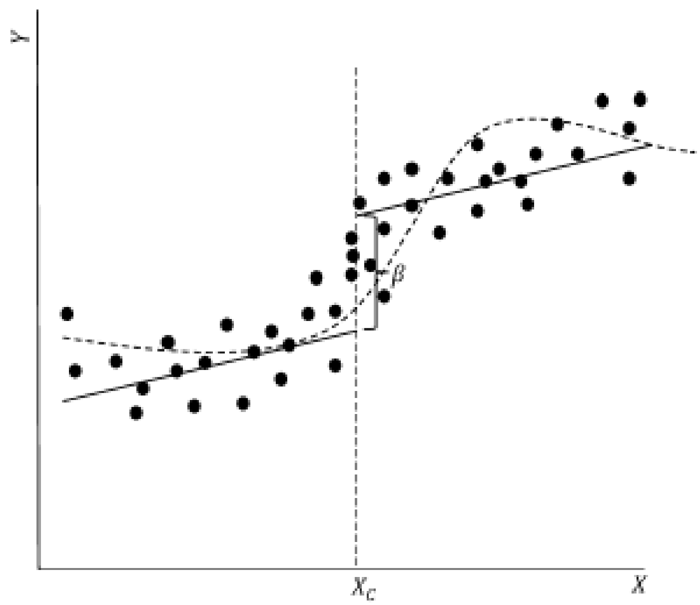


Figure 6: Misspecifying Regression Discontinuity Polynomial



Beware using a linear trend (solid line) for a non-linear tread (dashed line), can lead to misspecification (see Figure 6).

NOTE: rdplot is a nice command for plotting RDD graphics in Stata.
In any of the above designs, to capture the causal effect we require $E(\varepsilon_i|D_i, x_i) = 0$. As we have seen in other cases, this is an impossible assumption to test. However, it is common practice to check the balancing of covariates, i.e. the observable characteristics, such as: gender, age, past test scores are all balance, i.e. there are no statistical differences (t-tests) between those individuals just below and just above the cut-off point. The distribution of the running variable (x) should be smooth over the range of points close to the cut-off suggesting there has been no manipulation.
Regression discontinuity designs will fail if individuals can precisely manipulate the assignment variable. For example, if students could precisely choose their test score, through effort for example. Those who choose a score X^C or just above, will be systematically different from a student who chooses just below the cut-off.

Appendices

A Stata Commands for PSM

```
use "PSM.data.dta", clear
* t-tests for the 3 variables by t
ttest imrate, by(t)
ttest povrate, by(t)
ttest pcdocs, by(t)
* Probit model of t by covariates t determine probability of t=1
probit t povrate pcdocs
* Generate predictions of probability
predict prob, pr
* Identifying similar events and changing values
list imrate t povrate pcdocs prob
replace imrate=imrate[5] in 7/8
replace imrate=. in 9
replace povrate=povrate[5] in 7/8
replace povrate=. in 9
replace pcdocs=pcdocs[5] in 7/8
replace pcdocs=. in 9
replace prob=prob[5] in 7/8
replace prob=. in 9
* t-test of difference in means based on matched sample
ttest imrate, by(t)
* Reload data
use "PSM.data.dta", clear
* Estimating a probit mdel and then an ols model of selection
psmatch2 t povrate pcdocs, out(imrate)
* Testing matching
pstest povrate pcdocs
* Estimating a probit mdel and then an ols model of selection
psmatch2 t povrate pcdocs, out(imrate) common
* Testing matching
pstest povrate pcdocs
* Alternative procdure for PSM
teffects psmatch (imrate) (t povrate pcdocs, probit)
tebalance summarize
```

B R Commands for PSM

```
# Loading libraries
library(tidyverse)
library(foreign)
library(ggplot2)
library(haven)
library(readxl)
library(estimatr)
library(stargazer)
library(MatchIt)
library(lmtest)
library(sandwich)
# Loading dataset
psm <- read_dta("C:/Users/OneDrive - University of Warwick/PSM.data.dta")
# Difference in means tests for all variables by t
t.test(imrate ~ t, data = psm, var.equal = FALSE)
t.test(povrate ~ t, data = psm, var.equal = FALSE)
t.test(pcdocs ~ t, data = psm, var.equal = FALSE)
# Probit model of t vs covriates
model <- glm(t ~ povrate + pcdocs,
               data=psm, family = binomial(link = "probit"))
# Saving predicted probabilities
psm$fit1 <- fitted(model)
# Changing the predicted values of variables to match samples
psm$imrate[7] <- psm$imrate[5]
psm$imrate[8] <- psm$imrate[5]
psm$imrate[9] <- NA
psm$povrate[7] <- psm$povrate[5]
psm$povrate[8] <- psm$povrate[5]
psm$povrate[9] <- NA
psm$pcdocs[7] <- psm$pcdocs[5]
psm$pcdocs[8] <- psm$pcdocs[5]
psm$pcdocs[9] <- NA
psm$prob[7] <- psm$prob[5]
psm$prob[8] <- psm$prob[5]
psm$prob[9] <- NA
# Difference in means test on matched sample
t.test(imrate ~ t, data = psm, var.equal = FALSE)
# Rereading in data
```

```
psm <- read_dta("C:/Users/OneDrive - University of Warwick/PSM.data.dta")
# Using built in PSM function to estimate probit model
psm_step1 <- matchit(t ~ povrate + pcdocs,
                     data = psm, method = "nearest", distance ="glm",
                     ratio = 1,
                     replace = FALSE)
summary(psm_step1)
# Plotting the distribution of propensity scores
plot(psm_step1, type = "jitter", interactive = FALSE)
# Plotting the difference in means for original sample and matched sample
plot(summary(psm_step1), abs = FALSE)
# Plotting the raw and matched treated and control histograms
plot(psm_step1, type="hist")
# Estimating a difference in means
matched_data <- match.data(psm_step1)
psm_step2 <- lm(imrate ~ t, data = matched_data, weights = weights)
# Constructing clustered standard errors
coeftest(psm_step2, vcov. = vcovCL, cluster = ~subclass)
#Calculate the confidence intervals based on cluster robust standard error
coefci(psm_step2, vcov. = vcovCL, cluster = ~subclass, level = 0.95)
```

C Stata Commands for RDD

```
* Load data
use "RDD2.dta", clear
* A scatter plot of Y against x plotted for above and below cut-off
twoway scatter score_endyr score_startyr if( score_startyr<215), msymbol(oh) msize(small) /*
*/ || scatter score_endyr score_startyr if( score_startyr>=215), msymbol(of) msize(small)
* Storing the mean of score_endyr for score_strtyr below and above cut-off
su score_endyr if( score_startyr<215)
local mean1=r(mean)
su score_endyr if( score_startyr>=215)
local mean2=r(mean)
# Replotting scatter including mean values
twoway scatter score_endyr score_startyr if( score_startyr<215), msymbol(oh) msize(small) /*
*/ ||scatter score_endyr score_startyr if( score_startyr>=215), msymbol(of) msize(small) /*
*/ yline(‘mean1’, lcolor(blue)) yline(‘mean2’, lcolor(red))
* Difference in means test by cut-off dummy
ttest score_endyr, by(treat)
* Regression of score_endyr by d
reg score_endyr treat
* Aggregated scatter plot of score_startyr vs score_endyr
binscatter score_endyr score_startyr, rd(215)
* Using rdplot to achive same plot
rdplot score_endyr score_startyr, c(215)
* Difference in means for y, restricting the sample
ttest score_endyr if( score_startyr>=210 & score_startyr<=220) , by(treat)
* rdd regressions using quadratic, cubic and quartic terms
rdrobust score_endyr score_startyr, c(215) q(2)
rdrobust score_endyr score_startyr, c(215) q(3)
rdrobust score_endyr score_startyr, c(215) q(4)
* As above but including extra covariates
rdrobust score_endyr score_startyr, c(215) q(2) covs(age)
rdrobust score_endyr score_startyr, c(215) q(2) covs(age male)
rdrobust score_endyr score_startyr, c(215) q(2) covs(age male frlunch)
* Difference in means of covariates by running variable, restricting the sample
ttest age if(score_startyr>=210 & score_startyr<=220), by(treat)
ttest male if(score_startyr>=210 & score_startyr<=220), by(treat)
ttest frlunch if(score_startyr>=210 & score_startyr<=220), by(treat)
```

D R Commands for RDD

```
# Loading libraries
library(tidyverse)
library(foreign)
library(ggplot2)
library(haven)
library(readxl)
library(estimatr)
library(stargazer)
library(MatchIt)
library(lmtest)
library(sandwich)
library(rddtools)
# Loading dataset
rdd <- read_dta("C:/Users/OneDrive - University of Warwick/RDD2.dta")
# Creating a cut-off dummy variable
rdd <- rdd %>%
  mutate(d = ifelse(score_startyr<215, 0, 1),
         d = as.factor(d))
# Histogram of score_startyr looking for dicontinuity in running variable
ggplot(rdd, aes(x = score_startyr, fill = d)) +
  geom_histogram(binwidth = 1, color = "white", boundary = 215) +
  geom_vline(xintercept = 215) +
  labs(x = "Intial score", y = "Count", fill = "Above threshold")
# Scatter plot of score_startyr vs score_endyr by cut-off
ggplot(data=rdd, aes(x = score_startyr, y = score_endyr, color = d)) +
  geom_point() +
  geom_vline(xintercept = 215) +
  labs(y = "End of Year Score", x = "Statrt of Year Score")
# Calculating mea of score_endyr by d
mean <- rdd %>%
  group_by(d) %>%
  summarise(m_score_endyr=mean(score_endyr))
c1 <- mean[1,2]
c2 <- mean[2,2]
# Plotting mean values of score_endyr into plot
ggplot(data=rdd, aes(x = score_startyr, y = score_endyr, color = d)) +
  geom_point() +
  geom_vline(xintercept = 215.00) +
```

```
  geom_hline(yintercept = c(c1, c2), color = c("#F8766D", "#00BFC4")) +
  labs(y = "End of Year Score", x = "Statrt of Year Score") +
  scale_x_continuous(limits = c(175, 275)) +
  scale_y_continuous(limits = c(175, 275))
# T-test of score_endyr by treat
t.test(score_endyr ~ treat, data = rdd, var.equal = FALSE)
# Regression of score_endy by treat
reg_1 <- lm(score_endyr ~ treat, data = rdd)
# Scatter plot of score_endyr by score_startyr with regression lines
ggplot(data=rdd, aes(x = score_startyr, y = score_endyr, color = d)) +
  geom_point() +
  geom_smooth(method="lm") +
  geom_vline(xintercept = 215.0) +
  labs(y = "End of Year Score", x = "Statrt of Year Score")
# Simply regression of score_endyr by d and running variable interacted
rdd_lm1 <- lm(score_endyr ~ d*I(score_startyr-215), data = rdd)
summary(rdd_lm1)
# Creating an rdd data.frame
rdd_2 <- rdd_data(y = score_endyr, x = score_startyr, cutpoint = 215, data = rdd)
# Summarising and plotting variables
summary(rdd_2)
plot(rdd_2)
# rdd regression to mimic linear model above
rdd_lm2 <- rdd_reg_lm(rdd_object = rdd_2, order = 1)
summary(rdd_lm2)
# qunadratic rdd regression to mimic linear model above
rdd_lm3 <- rdd_reg_lm(rdd_object = rdd_2, order = 2)
summary(rdd_lm3)
# Creating vector of covariates
cvar1 <- data.frame(z1 = rdd$age, z2 = rdd$male, z3 = rdd$frlunch)
# Modying rdd data.frame to include covariates
rdd_3 <- rdd_data(y = score_endyr, x = score_startyr, covar = cvar1, cutpoint = 215, data =
# rdd regress with 1 covariate
rdd_lm4 <- rdd_reg_lm(rdd_object=rdd_3, order = 2, covariates = "z1")
summary(rdd_lm4)
# rdd regress with 2 covariates
rdd_lm5 <- rdd_reg_lm(rdd_object=rdd_3, order = 2, covariates = "z1 + z2")
summary(rdd_lm5)
# rdd regress with 3 covariates
```

```
rdd_lm5 <- rdd_reg_lm(rdd_object=rdd_3, order = 2, covariates = "z1 + z2 + z3")
summary(rdd_lm5)
# Modfying the sample based on a band-widthe kernel
bw_ik <- rdd_bw_ik(rdd_2)
reg_para_ik <- rdd_reg_lm(rdd_object=rdd_2, bw=bw_ik, order=2)
reg_para_ik
plot(reg_para_ik)
# T-test on a restricted ample
t.test(score_endyr ~ treat, data = subset(rdd, score_startyr>=210 & score_startyr<=220),
       var.equal = FALSE)
# T-test of covaraited on treatment
t.test(age ~ treat, data = subset(rdd, score_startyr>=210 & score_startyr<=220),
       var.equal = FALSE)
t.test(male ~ treat, data = subset(rdd, score_startyr>=210 & score_startyr<=220),
       var.equal = FALSE)
t.test(frlunch ~ treat, data = subset(rdd, score_startyr>=210 & score_startyr<=220),
       var.equal = FALSE)
```

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